

v4.0 Timing–Coverage Replay Evidence Report

Complete Findings, Tables, and Chart Compendium

run v4_0.timing.coverage.seed7_replay_20260701.083931 (full-evidence packet 20260701.111605)

Data sources (all inside the supplied archive): replay/summary.csv (12 candidate rows, 78 fields), replay/v4_0.nn.winners.csv (5 winner rows, now complete with winner_rank_within_trace and winner_selection), audit/normal.summary.csv (full $9 \times 5 = 45$ -row classical-prefetcher matrix), audit/evidence/trace_prefetch_evidence_report.md, audit/evidence/{trace_profile_summary, normal_prefetcher_evidence, standalone_prefetcher_evidence, normal_vs_standalone_evidence}.csv, audit/analysis/{normal_only_timely_standalone_reason, normal_vs_standalone_target_attribution, run_unique_event_outcomes, top_residual_contexts}.csv, train.metadata/v4_0_train.summary.csv (12 training rows incl. decision-ledger attribution), plan/v4_0_replay_contract.json, and both RUN_INFO.txt files. Every number below is computed directly from these files; no outside data is assumed. Where the report compares against v3.9, the v3.9 anchors are the figures from the previously supplied v3.9 packet.

Run identity and configuration (from RUN_INFO / MANIFEST): git commit a9738cd78ce87349e370f3abb53ac727fa3559e6; warmup 25,000,000 + simulate 25,000,000 instructions; replayer binary champsim.standalone_nn_replayer; normal binary perceptron-no-multi-no-ship-1core; chunk length 1024; export suffix pure.balanced.lru256; MAX_JOBS=2; normal matrix covers no_pref, stride, streamer, ampm, spp, ipcp, sms, sandbox, power7. The replay contract verifies expected_total=12 = actual 12 (602:4, 605:2, 619:3, 620:2, 623:1), with fresh_current_run_only=true and no_transformer=true. The audit driver reports [normal matrix PASS] 45 successful rows and [v4 replay re-parse PASS] 12 successful candidate rows.

1. Executive Summary

- v4.0 replays **12 candidates across 5 traces** under 5 named recipes: v31_assoc_abs_marginal_coverage (602, 4 variants), dependency_support_gated (605, 2), v31_direct_timing_calibration (619, 3), region_pair_marginal_coverage_union (620, 2), v33_context_lstm_retrained (623, 1).
- All 12 rows have replay_transport_ok=1, run_failed=0, drop_rate=0, keyed_dropped_invalid_address=0, keyed_unmatched_rows=0, and empty fail_reason: a clean run.
- Against the best classical prefetcher, v4.0 wins on the **same 2 of 5 traces as v3.9**: 605.mcf (+1.65% via support16_baseline) and 623.xalancbmk (+6.98% via context_lstm). 602.gcc (−1.63%), 619.lbm (−2.24%), 620.omnetpp (−1.90%) still trail.
- The headline change vs. v3.9 is **619.lbm**: explicit lead-aware issue policies raise the best IPC from 0.34651 to **0.37252 (+0.02601, +7.5% relative)**, cutting the late share from 13.05% to 2.14% (lead8) and the gap to sms from −9.06% to −2.24%.
- On the other four traces the v4.0 best is *at or slightly below* the v3.9 best: 602 0.43149 → 0.42918 (−0.00231), 605 0.19394 → 0.19185 (−0.00209), 620 0.24623 → 0.24227 (−0.00396), 623 0.37897 → 0.37863 (−0.00034, effectively a reproduction).
- Winner selection is now explicit in the data: winner_selection = max_replay_ipc_among_current_run_nn_candidates, winner_rank_within_trace=1 for all five winners. Notably, on 602 and 620 the winner is a current_run_control variant, i.e. the pre-declared provisional_primary *lost* to its own control on 2 of 5 traces.
- All 12 candidates still improve on the same-binary no-prefetch baseline, from +1.85% (620 marginal_union) to +16.63% (602 assoc_lru256/lru64).
- Cross-version caveat:** the same-binary no-prefetch IPC shifted between the v3.9 and v4.0 runs on every trace (602: 0.37585 → 0.36800; 605: 0.18798 → 0.18318; 619: 0.31911 → 0.32200; 620: 0.24187 → 0.23750; 623: 0.35341 → 0.35321), while best_normal_ipc is identical in both runs. Cross-version comparisons in this report therefore use replay_ipc and the fixed best_normal_ipc, not the speedup-vs-no-pref ratios.

Figure 1: IPC of all 12 v4.0 candidates vs no-prefetch and best classical prefetcher (green = beats best_normal; black outline = provisional_primary)

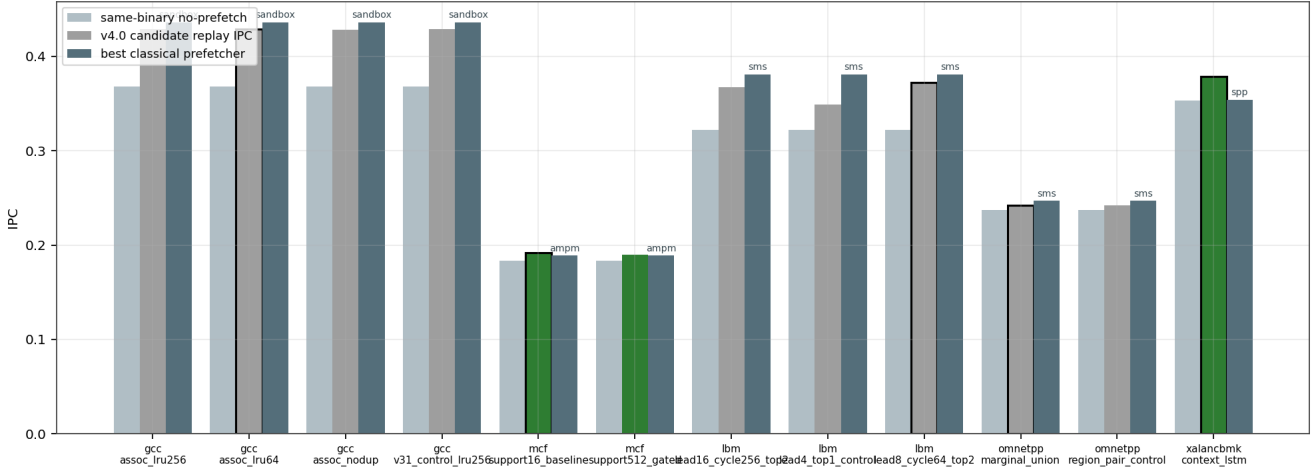
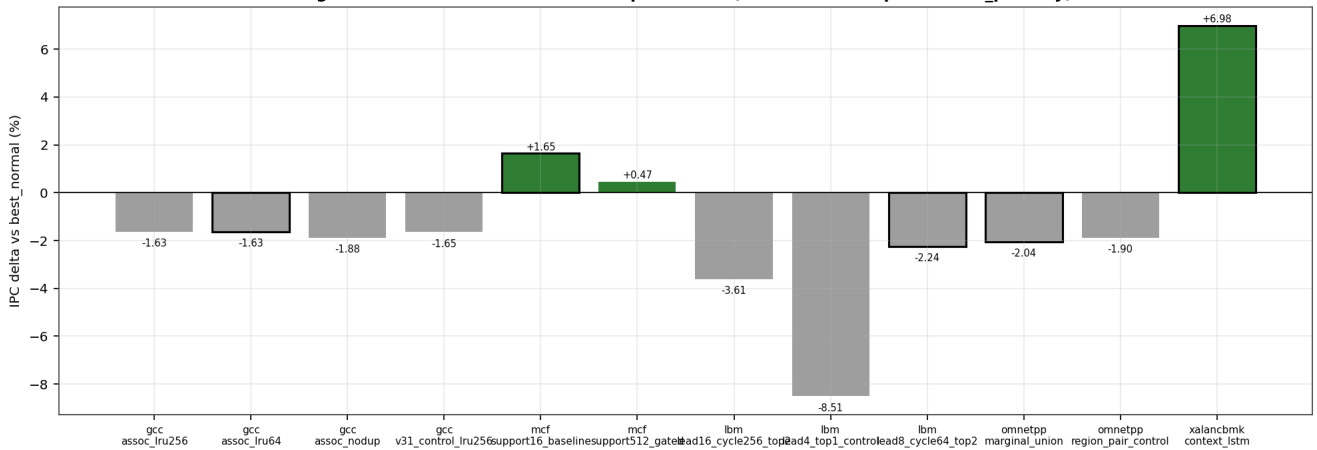


Figure 2: IPC delta vs best classical prefetcher (black outline = provisional_primary)



2. Complete v4.0 Replay Results (12 candidates)

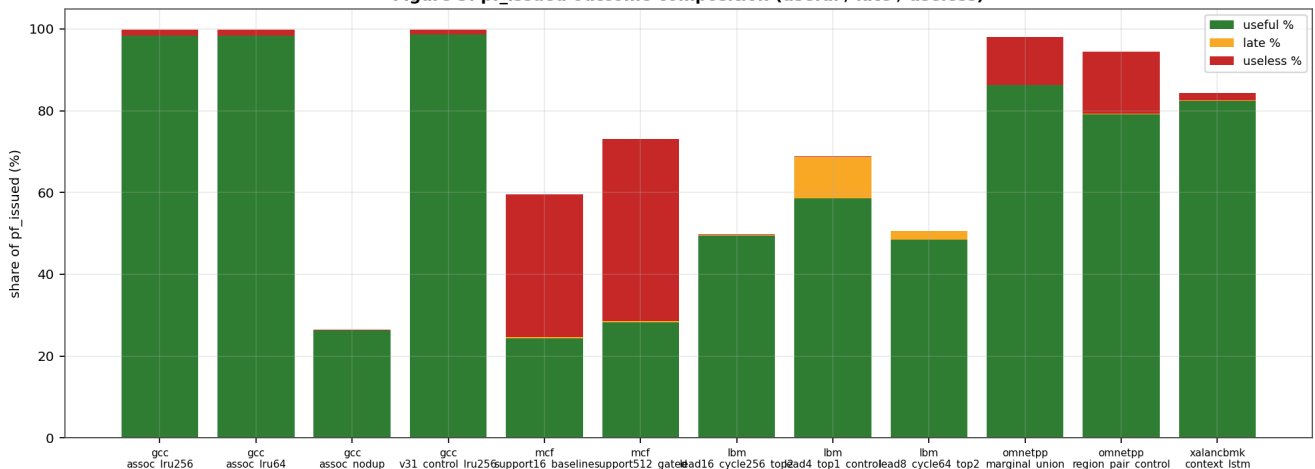
Table 1: Core replay results. *prim* = provisional_primary; *beats* = IPC > best_normal.

trace	candidate (policy_tag)	role	prim	replay_ipc	no_pref	bn_ipc	Δ bn %	beats
602.gcc	assoc_lru256	control	no	0.42918	0.36800	0.43628	-1.63	no
602.gcc	assoc_lru64	primary	yes	0.42918	0.36800	0.43628	-1.63	no
602.gcc	assoc_nodup	control	no	0.42806	0.36800	0.43628	-1.88	no
602.gcc	v31_control_lru256	control	no	0.42910	0.36800	0.43628	-1.65	no
605.mcf	support16_baseline	primary	yes	0.19185	0.18318	0.18874	+1.65	yes
605.mcf	support512_gated	control	no	0.18962	0.18318	0.18874	+0.47	yes
619.lbm	lead16_cycle256_top2	control	no	0.36729	0.32200	0.38105	-3.61	no
619.lbm	lead4_top1_control	control	no	0.34863	0.32200	0.38105	-8.51	no
619.lbm	lead8_cycle64_top2	primary	yes	0.37252	0.32200	0.38105	-2.24	no
620.omn	marginal_union	primary	yes	0.24190	0.23750	0.24695	-2.04	no
620.omn	region_pair_control	control	no	0.24227	0.23750	0.24695	-1.90	no
623.xal	context_lstm	primary	yes	0.37863	0.35321	0.35391	+6.98	yes

Table 2: Issue-outcome and quality fields for all 12 candidates.

trace	policy_tag	issued	useful	late	useless	useful%	late%	useless%	sel. acc
602	assoc_lru256	189,942	186,884	0	2,870	98.39	0.00	1.51	0.9839
602	assoc_lru64	190,021	186,884	0	2,868	98.35	0.00	1.51	0.9835
602	assoc_nodup	702,003	184,066	0	2,126	26.22	0.00	0.30	0.2622
602	v31_control_lru256	189,305	186,800	0	2,329	98.68	0.00	1.23	0.9868
605	support16_baseline	598,859	145,804	1,447	209,530	24.35	0.24	34.99	0.2435
605	support512_gated	448,018	126,686	991	199,982	28.28	0.22	44.64	0.2828
619	lead16_cycle256_top2	475,354	235,408	170	793	49.52	0.04	0.17	0.5769
619	lead4_top1_control	293,952	172,330	29,717	283	58.63	10.11	0.10	0.6170
619	lead8_cycle64_top2	485,695	235,205	10,377	129	48.43	2.14	0.03	0.5602
620	marginal_union	49,036	42,307	39	5,763	86.28	0.08	11.75	0.8628
620	region_pair_control	62,325	49,352	74	9,438	79.18	0.12	15.14	0.7918
623	context_lstm	482,470	397,393	695	9,112	82.37	0.14	1.89	0.8239

Figure 3: pf_issued outcome composition (useful / late / useless)



Findings from the transport / keyed fields (all 12 rows):

- keyed_direct_index_rows = keyed_entries = keyed_direct_index_rows_verified_pc_line on every row — every keyed entry is a verified direct-index (PC, line) row; keyed_mapped_cycle_pc_line_rows=0 everywhere.
- pf_issued = replayer_emitted_candidates on all 12 rows; pf_requested = pf_issued and pf_dropped = 0 on all 12 rows — the 1:1 emit-to-issue property of v3.9 is preserved.

- `keyed_trigger_coverage`: exactly 1.0 for all four 602 variants; 0.9931–0.9998 elsewhere; minimum 0.99314 (605 support512_gated).
- Duplicate trigger keys appear on 602 only, exactly as in v3.9: `keyed_unique_trigger_keys/keyed_entries` = 78.2% (lru256), 78.2% (lru64), 26.3% (nodup: 184,875/702,003), 80.1% (v31_control); all other traces are 100% unique.
- `replayer_runtime_l2_loads` again matches within a trace across variants (602: 404,219 all four; 605: 985,271/985,274; 619: 295,057/295,006/295,057; 620: 451,433/451,482; 623: 1,219,287), confirming variants replay the same demand stream.
- New field `pq_merged_duplicate_proxy` exposes in-flight duplicate merging: essentially 0 for the LRU-deduped exports (0–8, 115 on 623) but **65,867–67,326** for the 619 top-2 nodup policies and 14,628 for lead4 — the prefetch queue absorbs the duplicates that the exports no longer remove.
- New field `l2_load_miss_rate` of the replayed run: 602 ≈ 0.040 (vs 0.5025 no-pref), 619 lead8/lead16 ≈ 0.201 (vs 0.9987 no-pref), 605 ≈ 0.66 , 620 ≈ 0.60 , 623 0.051 (vs 0.3696 no-pref).

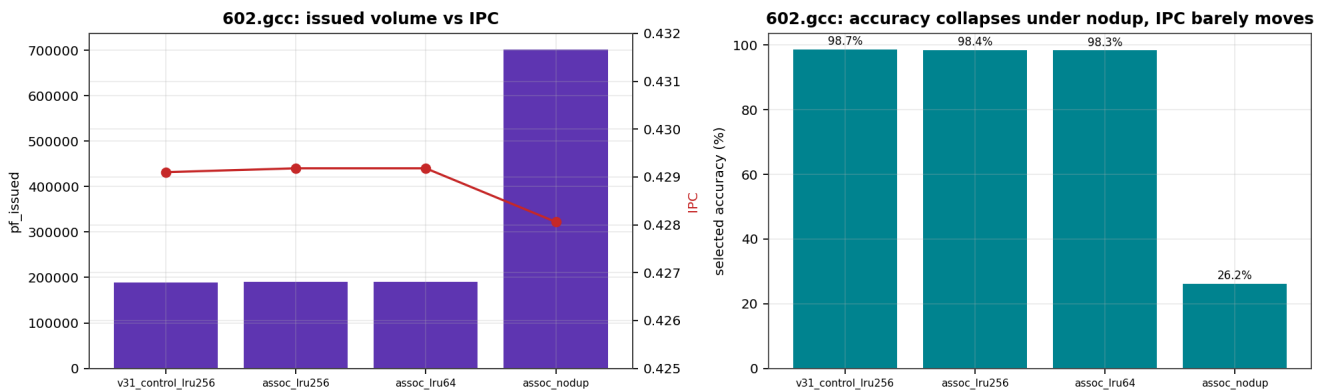
3. Per-Trace Experiments and What Each One Established

3.1 602.gcc — export-dedup capacity study (v31_assoc_abs_marginal_coverage)

Recipe rationale (verbatim from the plan): “reallocate fixed 64 V31 slots to include PC-current-line absolute successors, then replay calibrated LRU256/LRU64/no-LRU coverage variants.”

- **LRU256 vs LRU64 are indistinguishable:** identical IPC (0.42918), identical useful count (186,884), identical cycles (58,250,957); dedup capacity beyond 64 adds nothing on this trace.
- **Removing dedup entirely (nodup) multiplies issues by $3.7\times$** (189,942 $\rightarrow$ 702,003) while useful count barely moves (186,884 $\rightarrow$ 184,066) and IPC drops only 0.00112 (0.42918 $\rightarrow$ 0.42806). Selected accuracy collapses from 98.4% to 26.2% — demonstrating that *accuracy is a misleading metric under duplicated exports*: the same ~ 184 – 187 K useful prefetches are simply diluted by duplicate issues.
- The reallocated `assoc` bank does **not** beat the retrained `v31_hybrid_control` (0.42918 vs 0.42910, a 0.00008 difference), and both sit below the v3.9 best (0.43149) and below sandbox (0.43628).
- Coverage vs sandbox is unchanged from v3.9: unique demand-event coverage 92.0% (`lru256/lru64`), 90.6% (`nodup`), 92.0% (`v31_control`); `no_earlier_selected_standalone_export` still explains 99.6–99.8% of sandbox-only-timely events. The 602 gap remains a pure coverage gap that this bank reallocation did not close.

Figure 9: 602.gcc export-dedup capacity study (LRU256 / LRU64 / no-dedup)

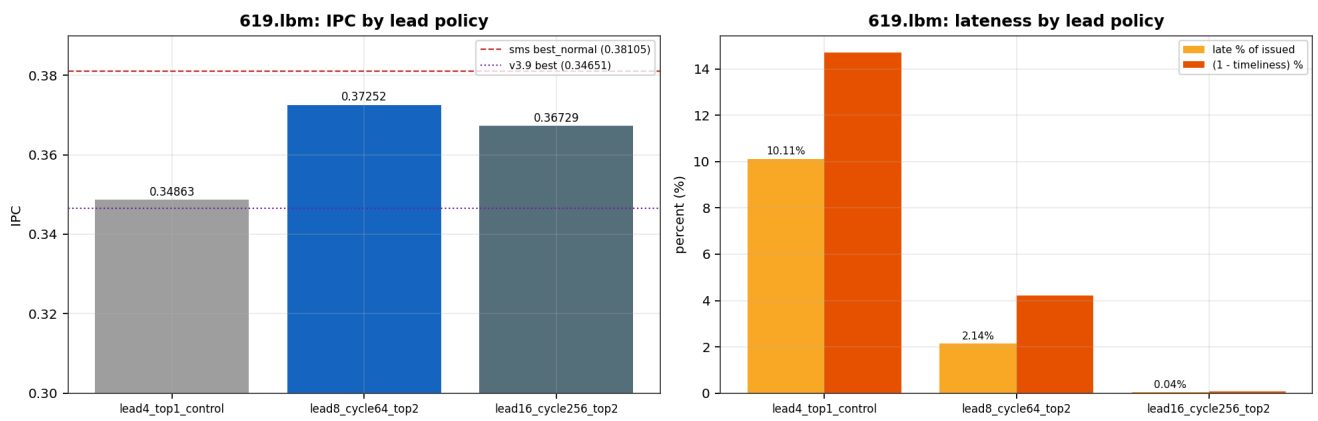


3.2 619.lbm — explicit timing calibration (v31_direct_timing_calibration)

Recipe rationale: “hold the V31 bank fixed and replay explicit early lead-aware top-2 policies rather than treating offline rank-0 coverage as a runtime timing solution.”

- **This is v4.0's biggest win.** `lead8_cycle64_top2` reaches IPC **0.37252** (v3.9 best: 0.34651; +0.02601, +7.5% relative), and `lead16_cycle256_top2` reaches 0.36729. The `lead4_top1` control (0.34863) closely reproduces the v3.9 operating point, isolating the causal factor: *the issue policy, not the model or bank*.
- Late share falls monotonically with lead: 10.11% (lead4) $\rightarrow$ 2.14% (lead8) $\rightarrow$ 0.04% (lead16); counter timeliness rises 0.853 $\rightarrow$ 0.958 $\rightarrow$ 0.999.
- **But maximum lead is not optimal:** lead16 has essentially zero lateness yet *lower* IPC than lead8 (0.36729 vs 0.37252) despite marginally fewer L2 load misses (59,235 vs 59,436). Issuing very early costs more cycles overall (68.07M vs 67.11M) — evidence of a timing sweet spot where excessive lead wastes bandwidth/occupancy even when “late” counters look perfect.
- Unique demand-event coverage jumps from 57.8% (v3.9 / lead4-like) to **79.7%** (lead8 and lead16): the top-2 policy also recovered coverage, not just timeliness.
- Overlap vs sms: both-timely rises 152,930 (v3.9) $\rightarrow$ 213,782 (lead8); sms-only-timely falls 102,340 $\rightarrow$ 41,488. Decomposition of the remaining sms-only events (lead8): 78.7% never exported (coverage), 20.7% exported-but-late (timing), so the residual -2.24% gap to sms is now roughly *4:1 coverage-dominated*, whereas the lead4 control splits 75.8%/23.9%.
- Ledger view (validation prefix): candidate-absent is only 201 of 59,008 events (0.3%) on 619 — the bank contains the targets; the loss channels are dedup/LRU suppression (20,475 for lead8's export) and policy rejection (8,701), consistent with timing/selection rather than representation being the binding constraint.

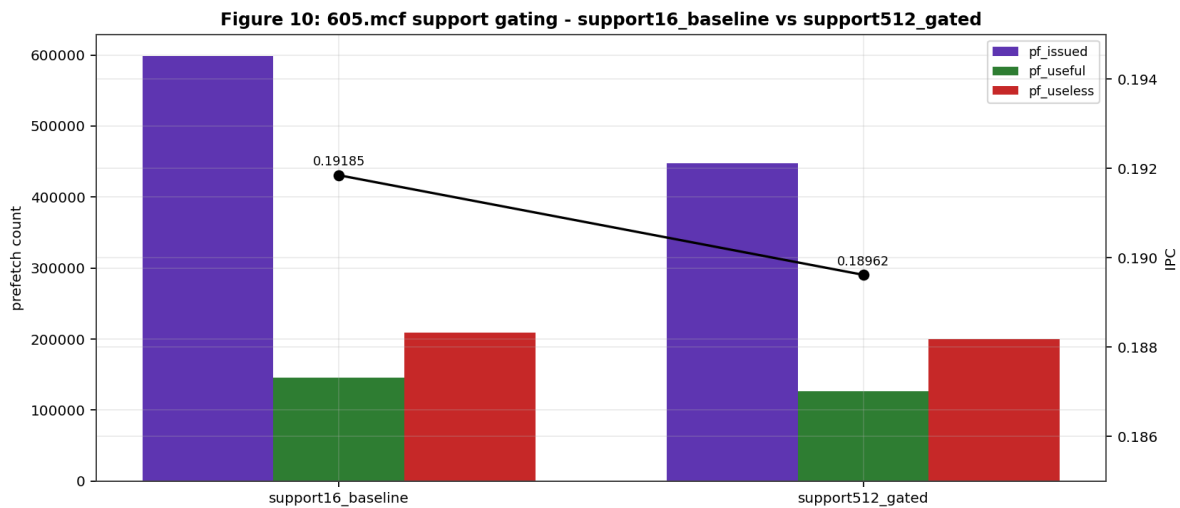
Figure 4: 619.lbm timing calibration - lead4 / lead8 / lead16



3.3 605.mcf — dependency support gating ([dependency_support_gated](#))

Recipe rationale: “retrain dependency baseline and a support-lower-bound gated dependency route; IPC chooses whether pollution reduction preserves coverage.”

- The gate works as designed on its own terms: support512_gated cuts issues 25% (598,859 → 448,018), cuts useless prefetches (209,530 → 199,982), and raises accuracy (24.3% → 28.3%).
- **But IPC answers “no”:** 0.18962 (gated) < 0.19185 (baseline). The gate removed useful coverage (useful 145,804 → 126,686; event coverage 18.4% → 16.1%) faster than it removed pollution. Both still beat ampm (+1.65% / +0.47%).
- Vs v3.9’s primary: v4.0 support16 issues fewer (598,859 vs 680,720), is more accurate (24.3% vs 19.2%), wastes less (35.0% vs 42.2% useless), covers more events (18.4% vs 16.4%) — yet its IPC is lower (0.19185 vs 0.19394). Every per-prefetch quality metric improved while the end-to-end result got slightly worse; the margin over ampm shrank from +2.76% to +1.65%.
- Ledger: **181,347 of 197,056 validation targets (92%) are candidate-absent** — the dependency banks simply do not contain most of what 605 needs. support512_producers=32 (preflight).
- Residual-context table: the standalone residual list is dominated by a single PC 0x40244f with delta = +1 spread across dozens of page offsets (71.6% of listed residual events) — a next-line pattern over highly diverse pages (605 has 102,356 unique load pages, by far the most of the five traces), which the current candidate representation does not capture.



3.4 620.omnetpp — marginal-coverage union ([region_pair_marginal_coverage_union](#))

Recipe rationale: “keep region-pair as control and construct a 64-slot region/V31/context marginal-coverage union from train-prefix candidate tables.”

- **The union lost to its own control:** marginal_union IPC 0.24190 vs region_pair_control 0.24227. The union issued fewer prefetches (49,036 vs 62,325), was more accurate (86.3% vs 79.2%), but covered fewer events (13.3% vs 15.4%) and delivered fewer useful prefetches (42,307 vs 49,352).
- Both v4.0 variants sit below the v3.9 primary (0.24623) and below sms (0.24695); the 620 gap (−1.9 to −2.0%) is still 98.9–99.2% “never exported” — an unresolved coverage problem on the most PC-diverse trace (6,591 unique PCs).
- Ledger: candidate-absent 63,995–65,683 of 90,280 (~72%), plus the largest model/policy-rejected counts outside 623 (16,617–19,099): both representation and selection lose targets here.

3.5 623.xalancbmk — retrained context LSTM ([v33_context_lstm_retrained](#))

Recipe rationale: “retrain only the proven v3.3 context-bank LSTM. No Transformer model, checkpoint, or replay candidate is created.”

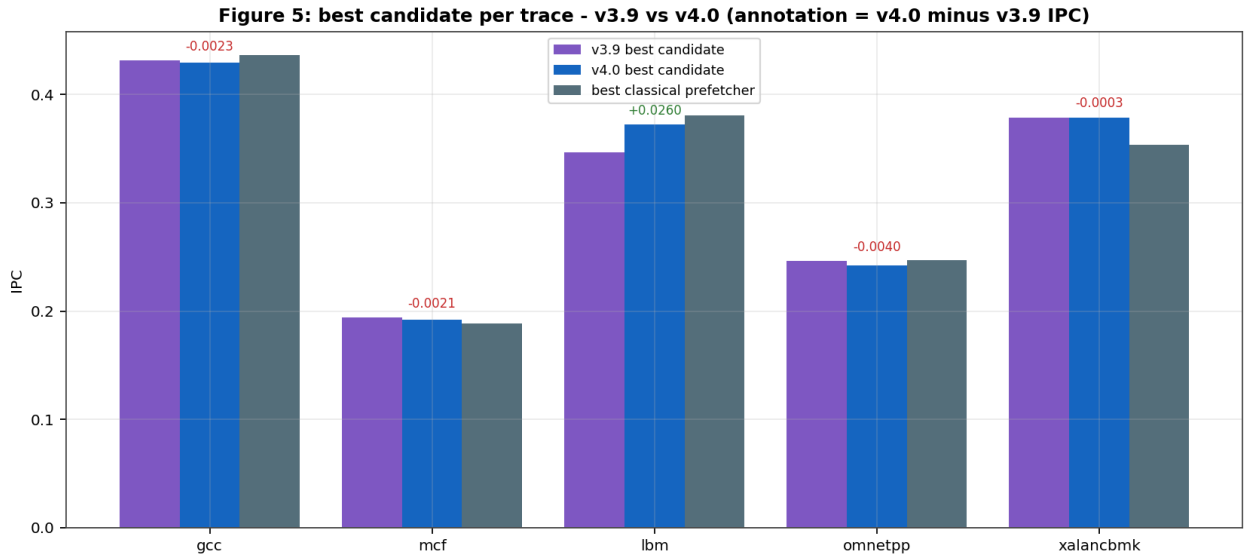
- **Clean reproduction:** 0.37863 vs v3.9’s 0.37897/0.37849 — the +6.98% margin over spp is a stable, re-trainable result, with near-identical overlap shape (standalone-only timely 388,429; both-timely 848; spp-only 284).

- spp remains a near-zero-value baseline here (selected accuracy 0.48%), and the wider matrix shows why 623 is special: *every aggressive classical prefetcher hurts this trace* (streamer -4.7% , sms -3.3% , sandbox -2.3% , power7 -2.8% , ampm -1.1% vs no-pref); only spp ($+0.20\%$) and stride ($+0.05\%$) are non-negative.
- Ledger: the dominant loss channel is **dedup/LRU suppression (103,296 targets)** — far larger than candidate-absent (34,430) or policy rejection (19,546) — yet IPC still wins, indicating heavy target duplication in time on this trace.

4. v3.9 → v4.0 Best-per-Trace Comparison

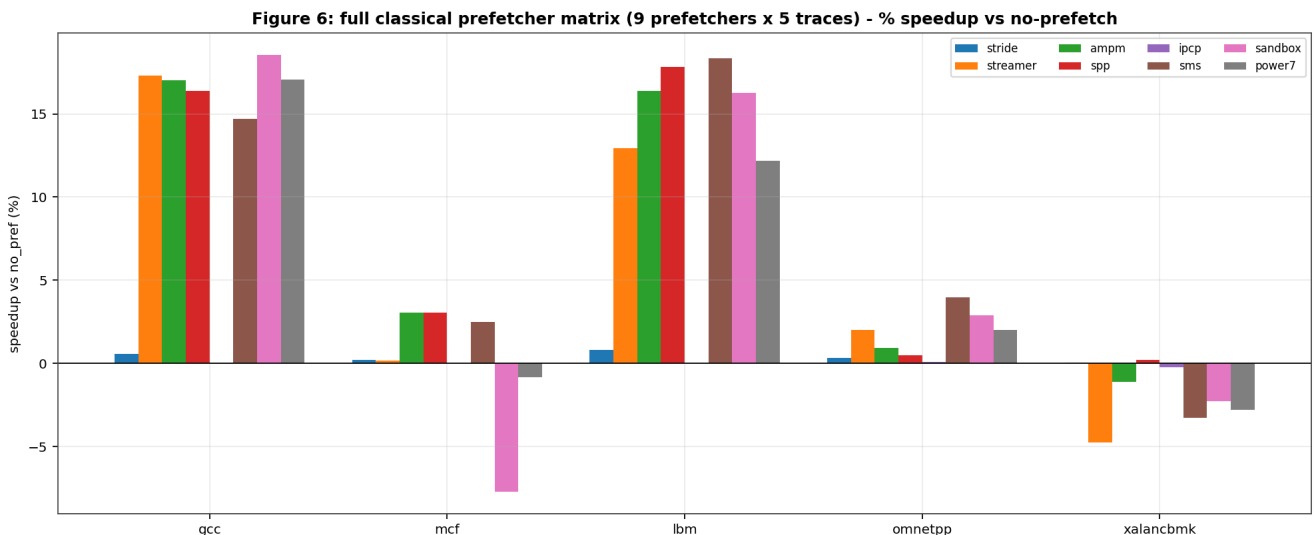
Table 3: Best candidate per trace, v3.9 vs v4.0 (best_normal_ipc identical across runs).

trace	v4.0 best candidate	bn	v3.9 IPC	v4.0 IPC	Δ	v3.9 vs bn	v4.0 vs bn
602.gcc	assoc_lru256/lru64	sandbox	0.43149	0.42918	-0.00231	-1.10%	-1.63%
605.mcf	support16_baseline	ampm	0.19394	0.19185	-0.00209	+2.76%	+1.65%
619.lbm	lead8_cycle64_top2	sms	0.34651	0.37252	+0.02601	-9.06%	-2.24%
620.omn	region_pair_control	sms	0.24623	0.24227	-0.00396	-0.29%	-1.90%
623.xal	context_lstm	spp	0.37897	0.37863	-0.00034	+7.08%	+6.98%



- Net: **one large gain** (619), **three small regressions** (602, 605, 620), one reproduction (623).
- The two “beats best_normal” traces are unchanged (605, 623); no trace flipped in either direction.
- 619’s coverage rose 57.8% → 79.7% and its late share fell 13.05% → 2.14%; it moved from the worst deficit (-9.06%) to third place (-2.24%), overtaking nothing but closing most of its own gap.

5. Full Classical-Prefetcher Matrix (new in v4.0: all 9 × 5 runs)



- Best classical per trace (unchanged): 602 sandbox 0.43628 (+18.55% vs no-pref); 619 sms 0.38105 (+18.34%); 605 ampm 0.18874 (+3.04%, tied by spp at 0.18874); 620 sms 0.24695 (+3.98%); 623 spp 0.35391 (+0.20%).
- **Classical prefetchers can actively hurt:** sandbox on 605 is -7.71% vs no-pref (IPC 0.16905, with 15.5M issues and 7.19M useless); on 623, streamer -4.75% , sms -3.27% , power7 -2.81% , sandbox -2.26% , ampm -1.12% , ipcp -0.23% . No v4.0 NN candidate is below its no-prefetch baseline on any trace.
- ipcp is effectively inert on all five traces (88–29,016 issues; speedup 0.9999–1.0009).
- On 605, ampm and spp tie on IPC (0.18874) with wildly different behavior (ampm: 509,840 issues at 17.8% accuracy; spp: 710,459 issues at 2.0% accuracy) — IPC equivalence does not imply mechanism equivalence.
- Event-level (from `run_unique_event_outcomes.csv`): sandbox on 602 covers 96.4% of no-pref misses with 99.98% timeliness-over-covered; sms on 619 covers 86.6% but only 91.0% timely-over-covered — while v4.619 lead16 achieves 99.93% timeliness-over-covered at 79.7% coverage.

6. Demand-Event Coverage Attribution (vs each trace's best classical prefetcher)

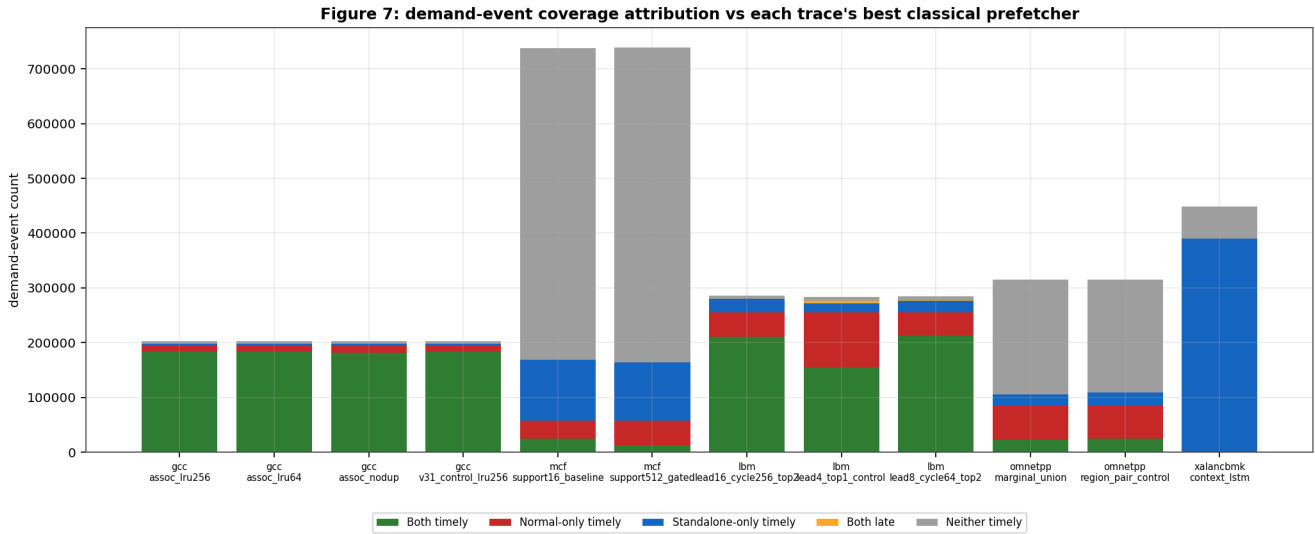
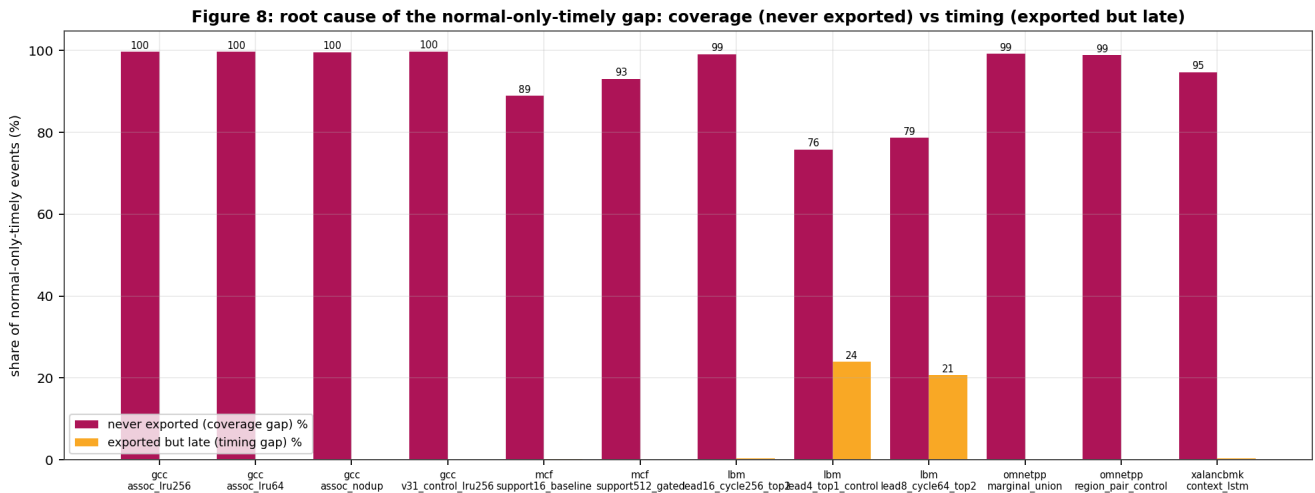


Table 4: Overlap vs best_normal and gap decomposition (all 12 candidates).

trace	variant	both	normal-only	alone-only	neither	no-export	sel-late	no-exp %
602	assoc_lru256	184,344	11,550	2,540	4,656	11,515	0	99.7
602	assoc_lru64	184,344	11,550	2,540	4,656	11,515	0	99.7
602	assoc_nodup	181,602	14,292	2,464	4,732	14,237	0	99.6
602	v31_control	184,232	11,662	2,568	4,628	11,636	0	99.8
605	support16	24,502	33,170	111,436	568,777	29,498	68	88.9
605	support512	12,806	44,866	106,208	574,411	41,768	36	93.1
619	lead16	210,612	44,658	24,266	6,550	44,264	135	99.1
619	lead4	155,197	100,073	16,921	7,439	75,840	23,962	75.8
619	lead8	213,782	41,488	20,959	7,203	32,631	8,597	78.7
620	marginal_union	21,968	64,039	20,036	208,806	63,530	14	99.2
620	region_pair	25,399	60,608	23,461	205,370	59,926	23	98.9
623	context_lstm	848	284	388,429	59,561	269	1	94.7



- The v3.9 conclusion generalizes to all 12 v4.0 candidates: the gap to the classical prefetcher is **coverage-dominated** (never-exported share 75.8–99.8%) everywhere; the only material exported-but-late shares are 619 lead4 (23.9%) and lead8 (20.7%).

- 623's disjoint-hit-set structure is intact: 388,429 standalone-only vs 848 both-timely — the NN candidate and spp cover nearly non-overlapping event sets.
- 605's "neither timely" mass remains enormous (568,777–574,411 of $\sim 740K$ no-pref misses, 77–78%): most of 605's misses are covered by neither approach.
- The analysis README explicitly cautions: "no_earlier_selected_standalone_export is not proof of candidate-bank absence" — the decision ledger (Section 8) was added to close exactly this gap.
- The full attribution table also covers all 8 non-best classical prefetchers per variant (108 rows total); e.g., against spp on 602, the v4 candidates are standalone-only timely on 57.9–59.3K events.

7. Winners Table (v4_0_nn_winners.csv — now complete)

Table 5: Per-trace winners. New columns `winner_rank_within_trace` (=1 for all) and `winner_selection` (= `max_replay_ipc_among_current_run_nn_candidates`) are present in the file.

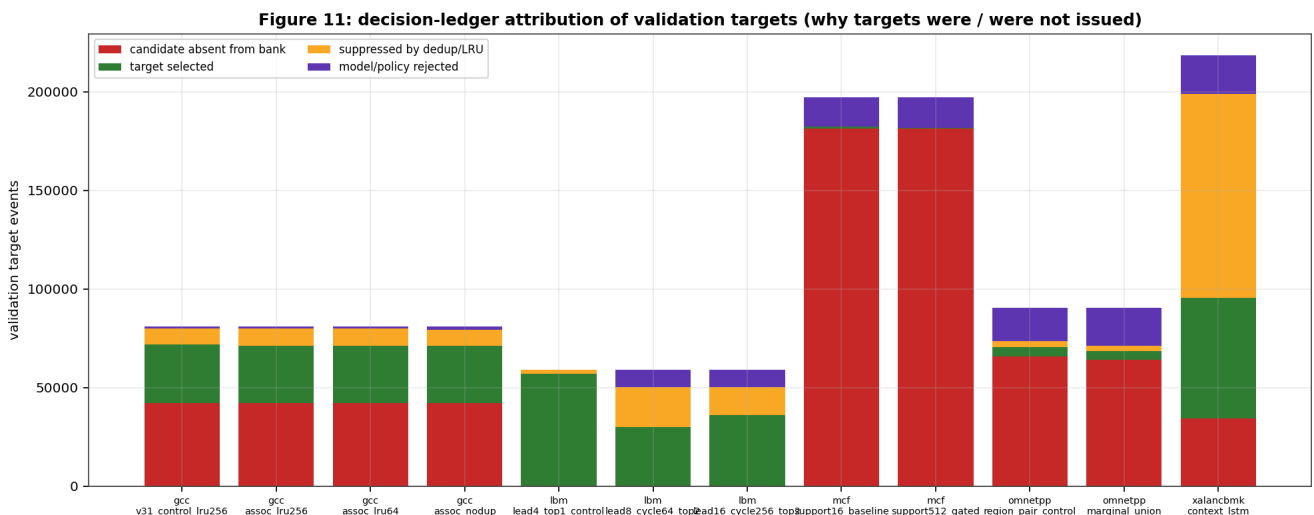
trace	winner	role	beats bn	IPC	cycles	sel. acc
602.gcc	assoc_lru256	control	no	0.42918	58,250,957	0.9839
605.mcf	support16_baseline	primary	yes	0.19185	130,308,565	0.2435
619.lbm	lead8_cycle64_top2	primary	no	0.37252	67,110,754	0.5602
620.omn	region_pair_control	control	no	0.24227	103,191,229	0.7918
623.xal	context_lstm	primary	yes	0.37863	66,027,345	0.8239

- On 602 the winner is `assoc_lru256` rather than the declared primary `assoc_lru64` — they tie on IPC to 5 decimals (0.42918) and on cycles (58,250,957); the tie-break favored the control. On 620 the control genuinely beat the primary (0.24227 > 0.24190). **The pre-registered primary lost or tied on 2 of 5 traces.**
- All five winners: `fail_reason` empty, `drop_rate`=0.0, `run_failed`=0.
- Cycle spread: 58.25M (602) to 130.31M (605), $\sim 2.24\times$, again tracking inverse IPC.

8. Training Metadata and Decision-Ledger Attribution (new evidence class in v4.0)

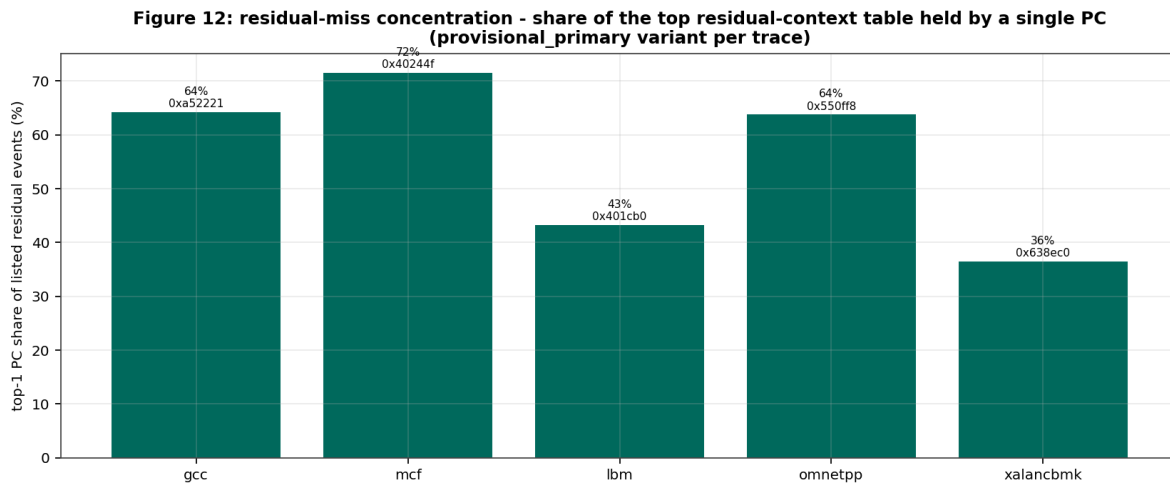
Table 6: Training summary (v4_0_train_summary.csv). All models: 4,147,959 parameters, `parity_max_abs_logit_diff`=0.0, gate status `pending_replay`.

trace	artifact / policy	val loss	val events	cand. absent	selected	dedup-suppr.	rejected
602	v31_hybrid / control	0.1108	80,844	42,200	29,636	8,078	925
602	assoc / lru256+lru64	0.1110	80,844	42,184	28,935	8,778	942
602	assoc / nodup	0.1110	80,844	42,184	28,825	8,312	1,518
619	timing / lead4	0.1740	59,008	201	56,710	2,085	7
619	timing / lead8	0.1740	59,008	201	29,626	20,475	8,701
619	timing / lead16	0.1740	59,008	201	35,825	14,180	8,797
605	support16	1.1369	197,056	181,347	1,079	0	14,626
605	support512	1.1141	197,056	181,347	424	1	15,280
620	region_pair control	0.7206	90,280	65,683	4,804	3,170	16,617
620	marginal_union	0.7163	90,280	63,995	4,351	2,829	19,099
623	context_lstm	0.3152	243,699	34,430	61,136	103,296	19,546



- Validation loss cleanly ranks trace difficulty: 602 (0.111) < 619 (0.174) < 623 (0.315) < 620 (0.716–0.721) < 605 (1.114–1.137) — and correlates with the IPC-vs-no-pref headroom each trace realized.
- The ledger finally separates *representation* from *policy*: 605 is a representation failure (92% candidate-absent); 619 is a policy/selection trade (0.3% absent; lead8 deliberately suppresses/rejects 29,176 targets vs lead4's 2,092 yet wins on IPC); 623 loses most targets to dedup/LRU suppression (103,296) and still wins — suppressed duplicates were redundant.
- Training wall-clock (per recipe): 619 1,387.7 s; 623 2,802.2 s; 620 3,148.5 s; 602 3,365.1 s; 605 4,360.5 s.
- Every parity check passed with zero max logit difference; all 12 rows share one architecture size (4.15M parameters), so cross-trace differences are attributable to data/recipe, not capacity.

9. Residual Contexts (now included — absent from the v3.9 packet)



- `top_residual_contexts.csv` ships 5,700 rows covering the 9 classical prefetchers and all 12 standalone variants (per trace: 602:1,300; 605:1,100; 619:1,200; 620:1,100; 623:1,000 rows).
- Residual misses are **highly PC-concentrated** for the provisional primaries: one PC holds 71.6% of 605's listed residual events (0x40244f, delta +1 across many page offsets), 64.2% of 602's (0xa52221), 63.8% of 620's (0x550ff8), 43.3% of 619's (0x401cb0, deltas 3/65/67), and 36.5% of 623's (0x638ec0).
- 623's residual deltas include enormous values (e.g. 1,600,961,158,017 and -4,189,474,377,000) — pointer-valued strides far outside any delta-table representation, plus a -2,008-line context; these are structurally different from the small-delta residuals of 602/605/619.
- 619's residuals sit on deltas 3, 65, 67 at page offsets 0–6 from one PC — a multi-stride stencil footprint; the same PC dominates all three lead policies, so the remaining 619 gap is one addressable pattern.

10. Consolidated Rankings and Totals

- By IPC delta vs best_normal (v4.0 winners): 623 (+6.98%) > 605 (+1.65%) > 602 (-1.63%) > 620 (-1.90%) > 619 (-2.24%). (v3.9 order was 623 > 605 > 620 > 602 > 619; 602 and 620 swapped, 619 closed most of its gap.)
- By IPC gain vs same-binary no-prefetch (v4.0 winners): 602 (+16.63%) > 619 (+15.69%) > 623 (+7.20%) > 605 (+4.73%) > 620 (+2.01%).
- By unique demand-event coverage (winners): 602 (92.0%) > 623 (86.4%) > 619 (79.7%) > 605 (18.4%) > 620 (15.4%).
- Totals across the 12 candidates: 3,766,980 prefetches issued; 2,743,119 useful (72.8%); 43,509 late (1.2%); 443,222 useless (11.8%).

Appendix A: Provenance and Known Limits

- Input paths (verbatim from the evidence report): trace profiles `roi25m.trace.profile_20260625_115128` (identical to v3.9 — same trace windows); normal summary and event attribution under `v4_0.timing.coverage.seed7_replay_20260701.083931.all_normals.20260701.092508`; standalone replay summary under `standalone_lstm_replay/v4_0.timing.coverage.seed7_replay_20260701.083931`.
- The audit `NN_VARIANTS` line also registers v3.1 and v3.3 artifact paths as reference variants of the explainability tool; no v3.x replay rows appear in this packet's summary — all 12 rows are `v4_0.fresh.current.run`.
- Analysis README limitation (verbatim): “Standalone exports contain selections only. `no_earlier_selected_standalone_export` is not proof of candidate-bank absence.” The decision ledger in `train.metadata` is the packet's own answer to that limitation.
- The event-level per-row CSVs (`*.events.csv.gz`) and the ledger candidate/event gz files are present in the archive but summarized here only through the packaged aggregate CSVs.

- Trace profile extras first exposed in this packet: unique load pages per trace — 602: 4,965; 619: 4,603; 605: **102,356**; 620: 32,774; 623: 2,203; taken-branch fraction: 619 is 89.6% taken (loop-dominated), 623 only 34.1%.

Every figure in this report was computed field-by-field from the supplied v4.0 evidence archive. The only external anchors are the v3.9 comparison numbers, which come from the previously supplied v3.9 packet.